

Part II

Sharp unrestricted even-power row coresets

The following 26 pages reproduce the preceding manuscript in full and without alteration. Its source and original archive are included alongside this document.

For fixed $p = 2s$, $s \geq 2$, its local Theorem 1.1 proves

$$S_{2s}(k, \varepsilon) = \tilde{\Theta}_s \left(\min \left\{ \frac{k^s}{\varepsilon^2}, \frac{k^{s+1/2}}{\varepsilon} + \frac{k^{s-1}}{\varepsilon^2} \right\} \right)$$

for every rank and accuracy, with arbitrary input rank and nonnegative original-row weights. Its complete upper and lower proofs are retained; it is not being cited as an independently verified external result.

How to read the historical scope notices. References in this part to a “remaining non-even gap” or to the whole entry being “partially resolved” describe the earlier manuscript’s scope. That previously missing case is addressed in Part I of the present combined manuscript. No claim of interpolation from even to non-even powers is used.

Numbering. Theorems, equations, sections, references, and page numbers in this part are local to the reproduced manuscript. The opening theorem and proof map in Part I specify how the two parts combine into the full result.

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